(1) What is trigonometry? Explain Its meaning and applications metioned in the chapter

Ans:
Trigonometry is branch of mathematics that studies relation between the sides and angle of a triangle.
The chapter explains the trigonometry can be used to find heighs and distance that are difficult to measure directly. such as height of monument.

(2) In a right triangle, define the six trigonometric ratios of an acute angle A.

Ans:
Consider right triangle, ABC, ~~define the six~~ right-anged at B, with respect

to angle A:
→ Hypotenious = AC
→ side opposite A = BC
→ Side adjacent A = AB

Therefor:
sin A = BC/AC
Cos A = AB/AC
tan A = BC/AB
Cosec A = AC/BC = 1/sin A
Sec A = AC/AB = 1/Cos A
Cot A = AB/BC = 1/tan A

Thus cosec, sec and Cot are the reciprocals of sin, cos and tan respectivly.